

Notes: Hanson, Shleifer, Stein, and Vishny (JFE, 2015)
Banks as Patient Fixed-Income Investors

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1 Big picture

- **Question.** What is the economic business model of traditional commercial banks when they compete with shadow banks? The paper asks why banks both issue stable deposit-like claims and hold long-term, relatively illiquid fixed-income assets.
- **Core idea.** Both traditional banks and shadow banks create safe “money-like” claims, but they do it with different funding technologies. Traditional banks use deposit insurance, equity capital, and sticky deposits to hold assets to maturity. Shadow banks use early-exit funding and collateral liquidation to keep their claims safe.
- **Main theoretical result.** Stable bank funding is valuable precisely when assets are safe in the long run but exposed to temporary market-price dislocations. Banks therefore have a comparative advantage in holding illiquid fixed-income assets with modest fundamental risk and meaningful transitory price volatility.
- **Asset-side implication.** Banks should not mainly hold Treasury bills. They should hold assets such as loans, agency MBS, CMOs, CMBS, ABS, municipal bonds, and other fixed-income assets that are relatively safe in expected cash-flow terms but less liquid in market-value terms.
- **Liability-side implication.** Banks’ stable funding is not a side detail. The liability side and asset side are complements: sleepy deposits make buy-and-hold investing possible, and buy-and-hold investing helps banks earn enough spread to cover the cost of operating a deposit franchise.
- **Historical evidence.** Before federal deposit insurance, U.S. commercial banks relied on deposits but held much more liquid and shorter-maturity assets. After deposit insurance, commercial banks could shift toward longer-term loans and securities.
- **Contemporary evidence.** Across fixed-income asset classes, banks hold larger market shares of more illiquid assets. Across intermediary types, institutions with stickier liabilities hold more illiquid asset portfolios.
- **Policy message.** Shadow banking can be too large because early-exit financing creates fire-sale externalities. Minimum haircuts on shadow-bank financing can push activity back toward stable banks when traditional banking is the socially better intermediation technology.
- **Economic takeaway.** Traditional banks are special because they are patient fixed-income investors. Their advantage is not simply lending expertise or deposit-taking in isolation, but the balance-sheet combination of stable money creation and illiquid, low-fundamental-risk asset holding.

2 Incremental contribution of the paper

- **From liability-centric banking to a full balance-sheet theory.** Prior work emphasizes that banks create safe money-like liabilities. This paper explains why those liabilities are naturally backed by illiquid fixed-income assets rather than only by safe liquid assets.
- **A theory of coexistence between banks and shadow banks.** The model does not simply say that one system dominates. It shows how traditional banks and shadow banks can coexist, with the equilibrium market share of each strategy depending on asset illiquidity and fundamental risk.

- **A comparative-advantage view of asset holding.** The paper gives a sharp mapping from asset characteristics to intermediary type. Traditional banks specialize in assets with low fundamental downside risk but high interim fire-sale risk; shadow banks specialize in more liquid assets.
- **A new interpretation of bank securities portfolios.** Banks' holdings of MBS, CMOs, CMBS, ABS, municipal bonds, and other non-Treasury securities are not merely liquidity buffers or pure moral-hazard risk-taking. They are part of a patient fixed-income investment strategy.
- **A historical contribution.** The paper links the introduction of deposit insurance to a structural shift in bank assets. Before stable insured deposits, banks behaved more like modern shadow banks: runnable funding forced them to hold short-term and liquid assets.
- **A contemporary measurement contribution.** The authors construct asset illiquidity, liability stickiness, and liability maturity indices using the Financial Accounts of the United States and Basel III-style liquidity categories.
- **A regulatory contribution.** The model shows why private shadow-bank haircuts can be too low and derives the logic of optimal regulatory haircuts as Pigouvian tools against fire-sale externalities.
- **A synthesis contribution.** The paper ties together deposit insurance, bank capital, asset illiquidity, shadow banking, market fire sales, and the empirical balance-sheet structure of financial intermediaries.

3 Data and measurement

3.1 Commercial bank balance-sheet facts

- The paper begins with stylized facts about U.S. commercial banks. Table 1 uses Call Report data for commercial banks with more than \$1 billion in assets as of year-end 2012.
- Bank liabilities are highly homogeneous. In 2012, the value-weighted average bank funded roughly 75.6% of assets with deposits and 11.4% with equity.
- Bank assets are more heterogeneous. Net loans account for about 51.7% of assets on average, while securities account for about 20.8%.
- The asset heterogeneity is especially important because banks with similar deposit funding can choose quite different mixes of loans and securities.

Interpretation: The paper starts from an empirical puzzle. Banks are consistently deposit-funded, but they do not simply hold cash or Treasuries. A theory of banking must explain the joint structure of the balance sheet.

3.2 Bank securities portfolios

- Bank securities holdings are not dominated by the most liquid assets. In 2012, Treasuries and agencies were small shares of bank securities portfolios.

- Mortgage-linked securities, including agency MBS, CMOs, and CMBS, make up a large fraction of bank securities holdings.
- This pattern is inconsistent with a pure precautionary-liquidity-buffer view of bank securities portfolios.
- The paper also estimates that a narrow-bank strategy of taking deposits and investing in T-bills would have earned only about 0.06% of deposits from 1984 to 2012, before allocating overhead costs.

Interpretation: Deposits are costly to gather and service. Banks therefore need assets that earn a spread over short-term safe rates. The observed securities portfolio looks like deliberate exposure to fixed-income spread risk and illiquidity, not merely liquidity management.

3.3 Historical data

- The historical evidence compares the modern deposit-insured banking system with the National Banking Era and early twentieth-century U.S. commercial banking.
- Before federal deposit insurance, banks still issued deposit-like claims, but they held much more cash, interbank deposits, and short-term loans.
- During panics, deposit withdrawals were met by reserve drawdowns, calling loans, and refusing to roll over maturing loans. This made bank assets behave like liquid collateral backing runnable funding.
- After the introduction of federal deposit insurance, banks were able to move into longer-term and less liquid lending and securities holdings.

Interpretation: The historical episode is important because it separates deposit-taking from stable insured funding. Deposit-taking alone does not produce modern banking. Stability changes the feasible asset side of the bank balance sheet.

3.4 Contemporary aggregate data

- For contemporary tests, the paper uses the Financial Accounts of the United States, formerly the Flow of Funds.
- The cross-section includes commercial banks, P&C insurers, life insurers, money market funds, GSEs, finance companies, REITs, and broker-dealers.
- The key data point is 2012Q4, although the paper says the qualitative findings are not highly sensitive to the date.
- Asset classes include Treasuries, GSE-backed securities, corporate bonds, municipal bonds, equities, mutual fund shares, mortgages, consumer loans, C&I loans, and security credit.

Interpretation: The empirical exercise is aggregate and cross-sectional. It asks whether the balance sheets of intermediary sectors line up with the model's comparative-advantage predictions.

3.5 Illiquidity, maturity, and stickiness indices

- The paper defines an asset illiquidity score $ILLIQUID_i \in [0, 1]$, where Treasuries are very liquid and loans are highly illiquid.
- It defines liability maturity $MATURITY_i \in [0, 1]$ based on contractual maturity.
- It defines liability stickiness $STICKY_i \in [0, 1]$ based on how runnable or withdrawal-sensitive a liability is.
- For intermediary type j , asset illiquidity is the asset-weighted average of $ILLIQUID_i$. Liability stickiness and maturity are liability-weighted averages of the corresponding liability scores.

Interpretation: Stickiness, not legal maturity alone, is the crucial concept for banks. Bank deposits are contractually short-term but can be behaviorally sticky because of insurance, relationships, and deposit-franchise frictions.

4 Models and empirical specifications

4.1 Household demand for money-like claims

The model has three dates, $t = 0, 1, 2$, risky assets indexed by i , and three sets of actors: households, traditional banks, and shadow banks. Households value both consumption and monetary services. A safe claim provides monetary services; a risky claim does not.

$$U = C_0 + \beta E[C_2] + \gamma M.$$

- M is the amount of perfectly safe money-like claims.
- β is the discount factor applied to risky future consumption.
- γ is the extra utility or convenience premium from holding safe claims.
- Safe claims are discounted at $1/(\beta + \gamma)$, while risky claims are discounted at $1/\beta$.

Interpretation: The convenience premium breaks Modigliani-Miller logic. The value of a risky asset depends on whether intermediaries can transform it into safe claims.

4.2 Asset payoffs and interim fire-sale risk

Each asset pays R at $t = 2$ in the good state and $z_i < R$ in the bad state. After pessimistic news at $t = 1$, the asset's fundamental value is

$$F_i = qR + (1 - q - \varepsilon)z_i.$$

However, if the asset must be sold at $t = 1$, the market price is only

$$k_i F_i \leq F_i.$$

- z_i captures fundamental safety. A higher z_i means lower downside cash-flow risk.

- k_i captures the fire-sale discount. A lower k_i means worse interim liquidation value.
- Asset illiquidity is indexed by ϕ_i . More illiquid assets have steeper fire-sale demand curves.

Interpretation: The key distinction is between fundamental risk and interim market-price risk. Traditional banks are useful when interim prices can temporarily fall below fundamental value.

4.3 Traditional banking strategy

Traditional banks use deposit insurance and equity capital to create stable deposits. They hold assets to maturity, so they do not need to liquidate at $t = 1$ after bad news.

The total value of claims a traditional bank can issue against asset i is

$$V_i^B = \gamma z_i + \beta[pR + (1 - p)F_i].$$

- The term $\beta[pR + (1 - p)F_i]$ is the risky discounted value of expected cash flows.
- The term γz_i is the money premium from using the asset to back safe insured deposits.
- Capital regulation and insurance premiums prevent traditional banks from simply exploiting deposit insurance.

Interpretation: Traditional banks create fewer money-like claims per dollar of collateral than shadow banks, but they avoid costly liquidation when prices temporarily fall.

4.4 Shadow banking strategy

Shadow banks create safe money-like claims through early exit. If pessimistic news arrives at $t = 1$, the shadow-bank chain liquidates collateral at the fire-sale price $k_i F_i$. This keeps the short-term claim safe but exposes the system to forced sales.

The total value of claims shadow banks can issue against asset i is

$$V_i^S(k_i) = \gamma k_i F_i + \beta[pR + (1 - p)k_i F_i].$$

- Shadow banks can create money-like claims up to the liquidation value of collateral, $k_i F_i$.
- The early-exit option makes short-term shadow-bank claims safe from the investor's perspective.
- The cost is unstable funding: if bad news arrives, liquidation occurs precisely when markets are weak.

Interpretation: Shadow banking is efficient for liquid collateral, but fragile for assets whose market value can fall sharply under forced sales.

4.5 Equilibrium between banks and shadow banks

Let μ_i be the fraction of asset i held by shadow banks. A higher μ_i increases the amount of forced liquidation after bad news and reduces the liquidation price $k(\mu_i, \phi_i)F_i$.

At an interior equilibrium where both traditional and shadow banks hold asset i , the two strategies break even:

$$V_i^B = V_i^S(k(\mu_i^*, \phi_i)).$$

The equilibrium condition can be written as

$$(1 - p)\beta [1 - k(\mu_i^*, \phi_i)]F_i = \gamma [k(\mu_i^*, \phi_i)F_i - z_i].$$

- Left side: marginal benefit of stable funding. Banks avoid fire-sale liquidation losses.
- Right side: marginal cost of stable funding. Banks create fewer money-like claims than shadow banks.
- The equilibrium fire-sale discount is

$$k_i^* = \frac{\gamma z_i + \beta(1 - p)F_i}{\gamma F_i + \beta(1 - p)F_i}.$$

Interpretation: The ownership mix adjusts until shadow banks' money-creation advantage is exactly offset by their expected fire-sale cost.

4.6 Comparative statics

- **More illiquid assets:** $\partial(1 - \mu_i^*)/\partial\phi_i > 0$. Traditional banks hold a larger share when fire-sale risk is more severe.
- **Safer fixed-income assets:** $\partial(1 - \mu_i^*)/\partial z_i > 0$. Traditional banks hold a larger share when downside cash-flow risk is lower.
- **Higher money premium:** $\partial(1 - \mu_i^*)/\partial\gamma < 0$. A stronger demand for money-like claims pushes more activity into shadow banking because shadow banks create more money-like claims per unit of collateral.
- **Better interim news environment:** $\partial(1 - \mu_i^*)/\partial p < 0$. When fire sales are less likely, stable funding is less valuable.

Interpretation: Banks should be most important in assets like mortgages, MBS, and other fixed-income claims that are safe enough to back deposits but illiquid enough to make early liquidation costly.

4.7 Empirical cross-section of asset classes

The first empirical test looks across asset types and asks whether banks hold larger shares of more illiquid assets.

$$BANK_SHR_i = \frac{A_{bank,i}}{\sum_j A_{ji}}.$$

The main univariate regression is

$$BANK_SHR_i = 0.04 + 0.57 ILLIQUID_i, \quad R^2 = 0.47.$$

With a crude control for fundamental safety, the regression becomes

$$BANK_SHR_i = 0.14 + 0.68 ILLIQUID_i + 0.16 FUNDSAFE_i, \quad R^2 = 0.73.$$

Interpretation: Banks' asset-class market shares rise strongly with illiquidity, consistent with the theory that stable funding is valuable for assets vulnerable to liquidation discounts.

4.8 Empirical cross-section of intermediary types

The second empirical test asks whether intermediary types with stickier liabilities hold more illiquid asset portfolios.

The liability-stickiness regression is

$$A_ILLIQUID_j = 0.13 + 0.55 L_STICKY_j, \quad R^2 = 0.64.$$

The liability-maturity regression is weaker:

$$A_ILLIQUID_j = 0.27 + 0.36 L_MATURITY_j, \quad R^2 = 0.29.$$

Interpretation: Stickiness matters more than contractual maturity. Banks are the key example: their liabilities are contractually short but behaviorally stable.

4.9 Optimal haircuts

The paper's regulatory section argues that shadow banks do not internalize their contribution to fire-sale discounts. A minimum haircut can reduce this externality by limiting money creation through unstable funding.

With no social cost of deposit insurance, the optimal haircut has the form

$$h_i^{**} = \frac{\eta(\phi_i)}{1 - \eta(\phi_i)} \frac{z_i}{F_i} > 0.$$

- $\eta(\phi_i)$ is the elasticity of the fire-sale discount with respect to liquidation volume.
- The haircut is higher for more illiquid assets.
- The haircut is also higher for assets with higher z_i , because these assets are especially suitable for stable traditional banking.

Interpretation: Regulation should lean most strongly against shadow-bank financing when the asset is both illiquid and safe enough that traditional banks can efficiently hold it.

5 Identification and empirical design

5.1 What the design can identify

- The paper combines a formal model with historical evidence and aggregate cross-sectional balance-sheet evidence.
- The empirical design identifies patterns consistent with the model: banks hold more illiquid fixed-income assets, and intermediaries with stickier liabilities hold less liquid portfolios.
- The cross-sectional regressions are descriptive tests of model predictions rather than causal micro estimates.
- The historical section provides suggestive before-and-after evidence around the introduction of deposit insurance.

Interpretation: The paper’s empirical strength is coherence across theory, history, and aggregate balance sheets. It is not a randomized or quasi-experimental design.

5.2 Main empirical predictions

- **Prediction 1.** Looking across assets and holding fundamental risk constant, banks should have a larger market share in more illiquid asset classes.
- **Prediction 2.** Looking across assets, more illiquid asset classes should be held by intermediaries with more stable funding.
- **Prediction 3.** Commercial bank portfolios should be more illiquid than shadow-bank portfolios, controlling for fundamental risk.
- **Prediction 4.** Across intermediaries, those with more stable funding should have more illiquid asset portfolios, controlling for fundamental risk.

Interpretation: These predictions translate the model’s μ_i , ϕ_i , and z_i objects into measurable market shares and portfolio characteristics.

5.3 Selection and omitted-variable concerns

- Asset illiquidity and fundamental risk may be correlated, so univariate regressions can suffer from omitted-variable bias.
- The paper partly addresses this by adding $FUNDSAFE_i$, based on risk weights, and by dropping equity-like instruments in robustness checks.
- The aggregate sample is small: the asset-class regressions have about 12 observations, and the intermediary-type regressions have about eight intermediary sectors.
- The model abstracts from diversification, monitoring, information production, and institutional boundaries, all of which may also affect asset holdings.

Interpretation: The empirical evidence should be read as disciplined stylized facts supporting the mechanism, not as definitive causal proof that funding structure alone determines asset portfolios.

5.4 Measurement advantages and limitations

- A strength is that the asset and liability indices are transparent and motivated by Basel III liquidity categories.
- Another strength is that the paper studies both sides of the balance sheet rather than treating bank assets or liabilities in isolation.
- A limitation is that the illiquidity, stickiness, and maturity scores require judgment when mapping Financial Accounts categories into economic concepts.
- Another limitation is that the Financial Accounts are aggregate. They do not reveal micro-level security selection, within-category liquidity, or bank-level funding differences.

Interpretation: The measurement is useful for matching broad theory to broad facts, but it necessarily compresses substantial heterogeneity inside each asset class and intermediary sector.

6 Tables and figures

6.1 Table 1: U.S. commercial bank balance-sheet composition, 2012

Read:

- Table 1 uses Call Report data for 501 U.S. commercial banking firms with assets above \$1 billion at year-end 2012.
- Panel A reports all banks. Panel B reports highly capitalized banks whose equity-to-assets ratio exceeds the industry median.
- The average bank is mostly deposit-funded: deposits equal 75.6% of assets in Panel A and 76.0% in Panel B.
- Equity is much smaller than deposits, at 11.4% in Panel A and 12.5% in Panel B.
- On the asset side, net loans are 51.7% of assets and securities are 20.8% for all banks. Banks also hold cash/reserves and other liquid assets.
- Securities holdings are not mainly Treasuries. MBS passthroughs, CMOs/CMBS, and other securities are much larger than Treasury holdings.

Economic message: The table establishes the motivating puzzle. Bank liabilities are deposit-heavy and homogeneous, but the asset side contains loans and longer-term securities rather than only safe liquid instruments.

Table 1

US commercial bank balance sheet composition, 2012.

This table illustrates the balance sheet composition of US commercial banks as of December 31, 2012 using Call Report data. We restrict attention to commercial banks with assets greater than \$1 billion. Collapsing all commercial banks owned by a single bank holding company into a single banking firm, our \$1 billion size cutoff leaves us with 501 banking firms as of year-end 2012. The table shows the value-weighted average balance sheet shares and (the equal-weighted) 90th and 10th percentiles of bank balance sheet shares. Panel A shows results for all banks. Panel B shows results for highly capitalized banks whose equity-to-assets ratio exceeds the industry median.

	Panel A: All banks			Panel B: Highly capitalized banks		
	VW Average	90th %-tile	10th %-tile	VW Average	90th %-tile	10th %-tile
Loans (gross)	52.9%	78.1%	42.2%	58.1%	78.1%	43.0%
Real estate	25.2%	62.0%	19.7%	27.4%	62.2%	17.6%
Residential	16.2%	26.1%	3.6%	18.1%	25.1%	2.9%
Commercial	9.0%	43.1%	9.7%	9.3%	43.8%	6.4%
C&I	9.7%	20.7%	3.0%	11.1%	23.4%	3.1%
Consumer	9.3%	9.2%	0.2%	10.4%	11.0%	0.1%
Other	8.7%	8.8%	0.1%	9.2%	8.5%	0.1%
Less reserves	-1.2%	-0.6%	-1.9%	-1.3%	-0.5%	-2.0%
Loans (net)	51.7%	75.9%	41.7%	56.8%	76.0%	42.4%
Liquid assets	35.2%	50.9%	16.4%	30.6%	49.1%	15.6%
Cash and reserves	10.2%	16.8%	2.2%	9.0%	14.1%	2.2%
Reverse repos	4.1%	1.4%	0.0%	2.2%	1.8%	0.0%
Securities	20.8%	40.7%	6.9%	19.3%	36.9%	6.5%
Treasuries	1.6%	1.4%	0.0%	1.9%	1.2%	0.0%
Agencies	1.2%	10.2%	0.0%	0.9%	7.7%	0.0%
MBS Passthroughs	6.8%	14.5%	0.2%	7.3%	14.5%	0.3%
CMOs and CMBS	5.2%	14.9%	0.0%	4.6%	13.3%	0.0%
Other securities	6.1%	13.4%	0.3%	4.6%	12.6%	0.2%
Trading assets (net)	3.8%	0.1%	0.0%	2.9%	0.1%	0.0%
Other assets	9.3%	11.6%	3.5%	9.7%	12.4%	3.9%
TOTAL ASSETS	100.0%			100.0%		
Deposits	75.6%	88.9%	73.6%	76.0%	86.0%	70.3%
Transaction	10.2%	22.6%	4.0%	9.5%	19.7%	2.0%
Savings	44.5%	63.0%	23.7%	48.3%	60.5%	23.6%
Time	9.4%	38.3%	8.6%	9.2%	39.1%	8.6%
Foreign	11.5%	0.0%	0.0%	9.1%	0.4%	0.0%
Repos	3.4%	6.2%	0.0%	2.6%	5.4%	0.0%
Subordinated debt	1.0%	0.5%	0.0%	0.9%	0.7%	0.0%
Other borrowed money	5.7%	9.4%	0.0%	5.4%	9.7%	0.0%
Other liabilities	2.9%	2.2%	0.3%	2.6%	2.8%	0.4%
Equity	11.4%	14.9%	8.2%	12.5%	16.2%	10.6%
TOTAL LIABILITIES & EQUITY	100.0%			100.0%		

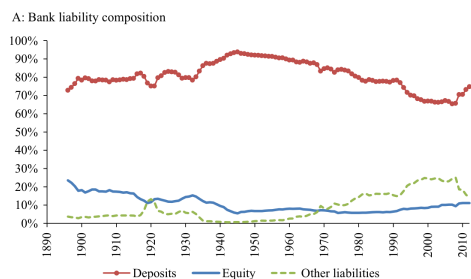
Table 1: U.S. commercial bank balance-sheet composition, 2012.

6.2 Figure 1: Data on U.S. commercial bank balance sheets, 1896–2012

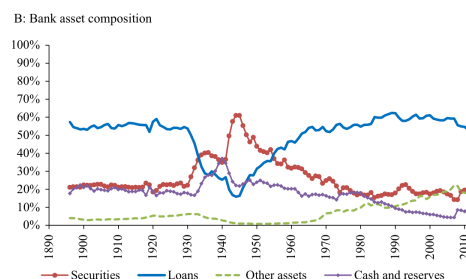
Read:

- Panel A shows that deposits have been the dominant source of bank funding for more than a century.
- Liability composition is relatively stable over time, especially compared with nonfinancial firms.
- Panel B shows much more movement on the asset side. The mix of cash/reserves, securities, loans, and other assets changes over time.
- The historical movements support the idea that the asset side is where the bank business model adapts when funding stability changes.

Economic message: The figure shows the asset-liability contrast at the heart of the paper. Banks are consistently deposit-funded, but their assets evolve with the institutional environment.



(a) Panel A: Bank liability composition.



(b) Panel B: Bank asset composition.

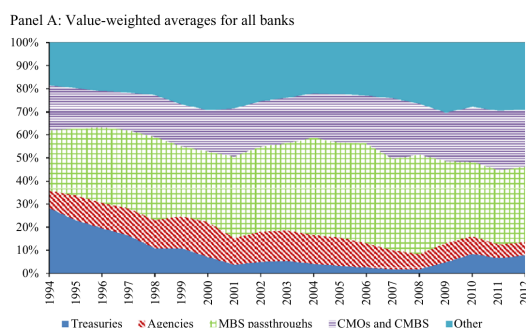
Figure 1: U.S. commercial bank balance sheets, 1896–2012.

6.3 Figure 2: Composition of bank securities portfolios, 1994–2012

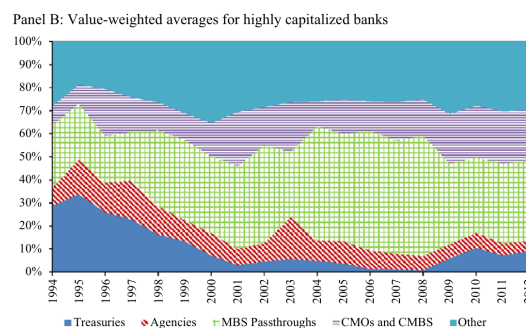
Read:

- Panel A reports the value-weighted average securities portfolio for all banks above \$1 billion in assets.
- Panel B repeats the exercise for highly capitalized banks.
- Both panels show that banks hold relatively little in Treasuries and agencies compared with MBS passthroughs, CMOs/CMBS, and other securities.
- The similarity between all banks and highly capitalized banks weakens a simple moral-hazard explanation based only on deposit insurance.

Economic message: Bank securities portfolios look like spread-earning fixed-income portfolios, not like pure liquidity buffers. This supports the model’s claim that banks specialize in assets that are low in fundamental risk but exposed to price and liquidity risk.



(a) Panel A: All banks.



(b) Panel B: Highly capitalized banks.

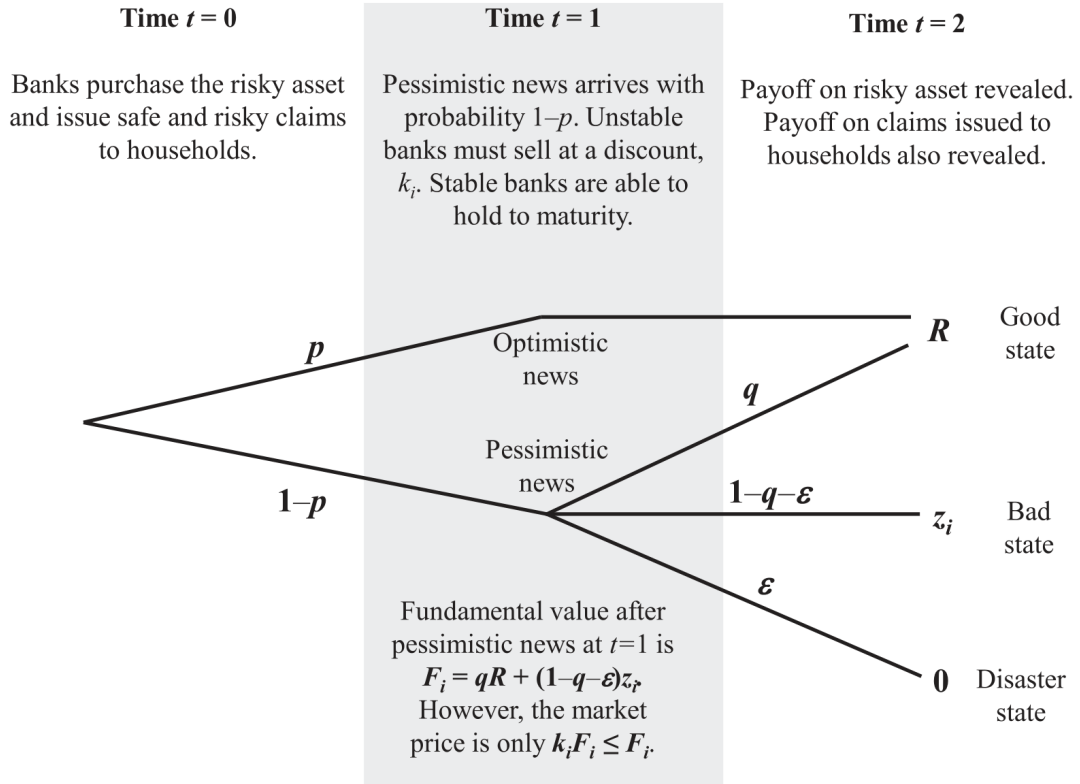
Figure 2: Composition of bank securities portfolios, 1994–2012.

6.4 Model timing diagram

Read:

- At $t = 0$, banks and shadow banks purchase risky assets and issue safe and risky claims to households.
- At $t = 1$, pessimistic news may arrive. Shadow banks must sell at a discount k_i , while stable banks can hold to maturity.
- At $t = 2$, asset payoffs are realized: a good state with payoff R , a bad state with payoff z_i , or a disaster state with payoff zero.
- After pessimistic news, fundamental value is $F_i = qR + (1 - q - \varepsilon)z_i$, but the market price can be only $k_i F_i \leq F_i$.

Economic message: The diagram summarizes the paper's central friction. The problem is not just asset risk; it is the interaction of interim liquidation needs with transitory market discounts.



Model timing and payoff tree.

6.5 Figure 3: The cross-section of asset classes

Read:

- Panel A plots commercial banks' market share against the asset illiquidity index for major financial asset classes.
- The slope is positive. Banks hold little of Treasuries but large shares of home mortgages, commercial mortgages, multifamily mortgages, consumer loans, and C&I loans.

- Panel B generalizes from banks to the average liability stickiness of asset holders.
- More illiquid assets are held by intermediaries whose liabilities are stickier on average.

Economic message: Figure 3 is the main asset-class evidence. It shows that the bank share of an asset class rises with illiquidity, consistent with the comparative advantage of stable funding.

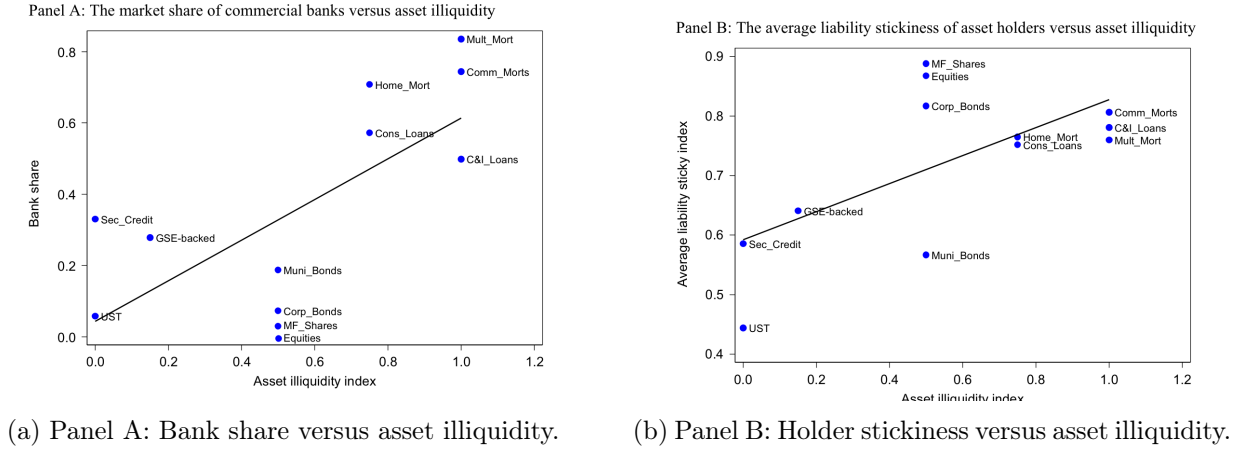


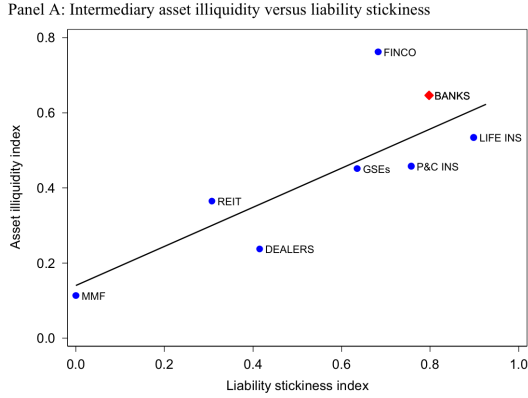
Figure 3: The cross-section of asset classes.

6.6 Figure 4: The cross-section of intermediary types

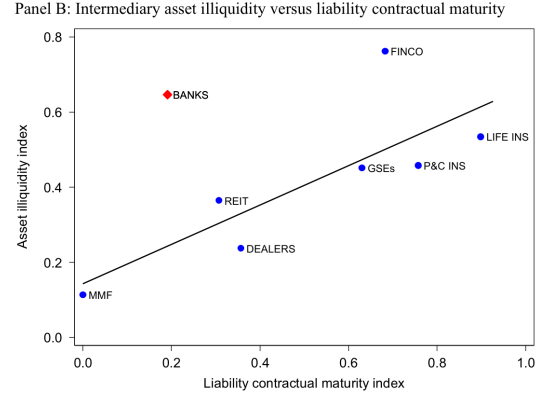
Read:

- Panel A plots intermediary asset illiquidity against liability stickiness.
- The relationship is strongly positive. Banks appear as institutions with sticky liabilities and illiquid assets.
- Panel B plots asset illiquidity against liability contractual maturity.
- The relationship is weaker because banks have contractually short-term liabilities that are nevertheless behaviorally sticky.

Economic message: Figure 4 shows why stickiness is the right funding concept. Contractual maturity misses the essence of bank deposits: they are short-term claims, but deposit insurance and deposit-franchise frictions make them stable.



(a) Panel A: Asset illiquidity versus liability stickiness.



(b) Panel B: Asset illiquidity versus liability maturity.

Figure 4: The cross-section of intermediary types.

7 Short summary

- The paper explains the business model of traditional commercial banks in a financial system that also contains shadow banks.
- Both bank and shadow-bank systems create safe money-like claims, but traditional banks do so with stable insured deposits and equity capital, while shadow banks do so with early-exit funding and collateral liquidation.
- Stable funding lets traditional banks avoid selling assets at temporary fire-sale discounts.
- The cost of stable funding is that banks create fewer money-like claims per dollar of collateral and bear the costs of capital, insurance, regulation, and deposit-franchise operations.
- Therefore, banks have a comparative advantage in assets that are safe in long-run cash-flow terms but illiquid and volatile in interim market-value terms.
- The theory explains why banks hold loans, MBS, CMOs, CMBS, ABS, and municipal/corporate fixed-income assets, but hold relatively little in Treasuries or equities.
- Historically, before deposit insurance, U.S. banks held much shorter and more liquid assets, resembling modern shadow banks.
- In contemporary aggregate data, banks' market share is higher in more illiquid asset classes, and intermediaries with stickier liabilities hold more illiquid asset portfolios.
- Shadow banking can be excessive because each shadow bank does not internalize how its liquidation worsens fire-sale prices for others.
- Regulatory haircuts can reduce the fire-sale externality by making shadow-bank funding more expensive for assets that are illiquid and suitable for bank holding.

8 Conclusion

8.1 Core conclusion

Hanson, Shleifer, Stein, and Vishny argue that the specialness of traditional banks lies in the combination of stable money creation and patient fixed-income investing. Banks are not merely delegated monitors, and they are not simply narrow issuers of deposits backed by Treasury bills. Their business model is to issue stable, sleepy liabilities and use the resulting patience to hold assets whose long-run cash flows are relatively safe but whose market prices can be temporarily depressed.

8.2 Economic intuition

The intuition is that funding stability is a form of investment horizon. A shadow bank must sell when short-term investors exit, so it needs collateral with high liquidation value. A traditional bank, protected by deposit insurance and supported by equity capital, can wait. This patience is especially valuable for fixed-income assets whose fundamental payoffs are not very risky but whose interim prices can deviate from fundamentals because of fire sales, noise-trader shocks, limited arbitrage capacity, or adverse selection.

8.3 Incremental contribution revisited

The paper advances the literature by linking the asset and liability sides of bank balance sheets. Liability-centric theories explain why banks create deposits, but not why deposits are backed by illiquid loans and securities. Asset-centric theories explain bank lending but not why banks are funded by deposits. This paper shows that the two sides are complements: stable liabilities make illiquid fixed-income assets attractive, and illiquid fixed-income assets help banks earn enough return to support costly deposit franchises.

8.4 Implications for banking and shadow banking

The model implies that migration from traditional banks to shadow banks is not always efficient. When shadow banks fund illiquid assets with runnable claims, they create fire-sale externalities. The same asset may be safer socially when held by a stable bank that can ride out temporary price dislocations. However, traditional banking also has social costs, especially because deposit insurance exposes taxpayers to tail risk. Optimal regulation must balance fire-sale externalities in shadow banking against the fiscal and moral-hazard costs of insured banking.

8.5 Limitations

The paper's empirical tests are aggregate and descriptive. The cross-sections are small, the asset/liability indices require judgment, and the model abstracts from diversification, monitoring, bank-level heterogeneity, and institutional boundaries. The theory also treats money-like claims as perfect substitutes and simplifies the structure of shadow banking chains. These simplifications are useful for clarity but limit direct quantitative interpretation.

8.6 Takeaway

The enduring takeaway is that banks are patient fixed-income investors. Their stable liabilities are valuable not because all bank assets are perfectly safe, but because many fixed-income assets are fundamentally safe enough to back money-like claims while still being illiquid enough that forced liquidation is costly. The modern commercial bank is therefore best understood as a bridge between households who want safe claims they do not need to monitor and markets where even low-fundamental-risk assets can have volatile, temporarily depressed prices.